Università della Svizzera italiana — Faculty of Economics
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FINANCIAL ECONOMETRICS – Homework 1

Due date: 23:59, October 19th, 2026

SIMULATION PRACTICE: SIMPLE OLS REGRESSION

In this homework, you are required to conduct a simulation on your own to better understand the Ordinary Least Square (OLS) regression model. This homework consists of (1) producing a short PDF report (using LATEX or Word) that contains numerical results and your interpretation of results and (2) a separate file containing the programming code (R or MATLAB or Python) that you used to produce the results.

The univariate OLS regression model

Consider the model function

$$y_i = \beta_0 + \beta_1 x_i \tag{1}$$

which describes a line with slope β_1 and intercept β_0. In general such a relationship may not hold exactly for most values of the independent and dependent variables; we call the unobserved deviations from the above equation the errors.

Suppose we observe n data pairs and call them $\{(x_i, y_i), i = 1, ..., n\}$. We can describe the underlying relationship between y_i and x_i involving this error term ε_i by

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \text{for } i = 1, ..., n \tag{2}$$

The equation 2 is called a linear regression model.

1. **Generate your data points**

   x_i is your independent variable which follows a uniform distribution $U(0,1)$.Your dependent variable y_i is a linear function of x_i as in equation (2) with $\beta_0 = 5$ and $\beta_1 = -2.5$. Due to some measurement deficiency, y_i is always contaminated with an error term ε_i, which is normally distributed as $N(0, V)$.

   - Suppose the variance of your error term $V = 0.16$, use the information above to generate $n = 20$ observations of data pairs (x_i, y_i).
   - Store your observations of y_i, x_i and ε_i into three $n \times 1$ vectors, and name them as y, x and ε.

2. **Run OLS regression**

   We are able to write equation (2) in matrix form as:

$$y = X\beta + \varepsilon, \tag{3}$$

   where $X = [\mathbf{1} \quad x]$ is an $n \times 2$ matrix ($\mathbf{1}$ represents a vector of ones), $\beta = [\beta_0 \quad \beta_1]'$ is a 2×1 vector, y, x and ε are all $n \times 1$ vectors. Since our model will usually contain a constant term, one of the columns in the X matrix will contain only ones.

   Now that you have generated your sample data, take the perspective of an econometrician who does not know the value of β, run an OLS regression of y on X following the steps in below.

   - Construct matrix X, then obtain your estimated parameter $\hat{\beta}$ by using the OLS estimator

$$\hat{\beta} = (X'X)^{-1}X'y \tag{4}$$

- Report your estimates $\hat{\beta}$.

3. **Plot your results**

   In order to better understand how well the OLS regression performs, plot the values of y_i versus x_i as scattered points and the estimated linear model $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ as a straight line in the same figure with x as your horizontal axis.

4. **Alternative data generating processes**

   Let's call the data generating process that you just conducted *DGP1*, where $V = 0.16$ and $n = 20$. Now, repeat all the steps above, plot your figures and report your results for the following two different data generating processes:

   - DGP2: $V = 1$ and $n = 20$
   - DGP3: $V = 1$ and $n = 200$

   Run your code 10 times (to generate different random variables) and calculate the average of your estimates β for each DGP. Compare your results for different DGPs. What do you find? Why do you think it's the case?